PaddyGuard AI:

A Multimodal AI Based System for Rice Leaf Disease And Pest Detection, Treatment Recommendation And Knowledge Assistance



Project ID : R26-SE-015

Introduction

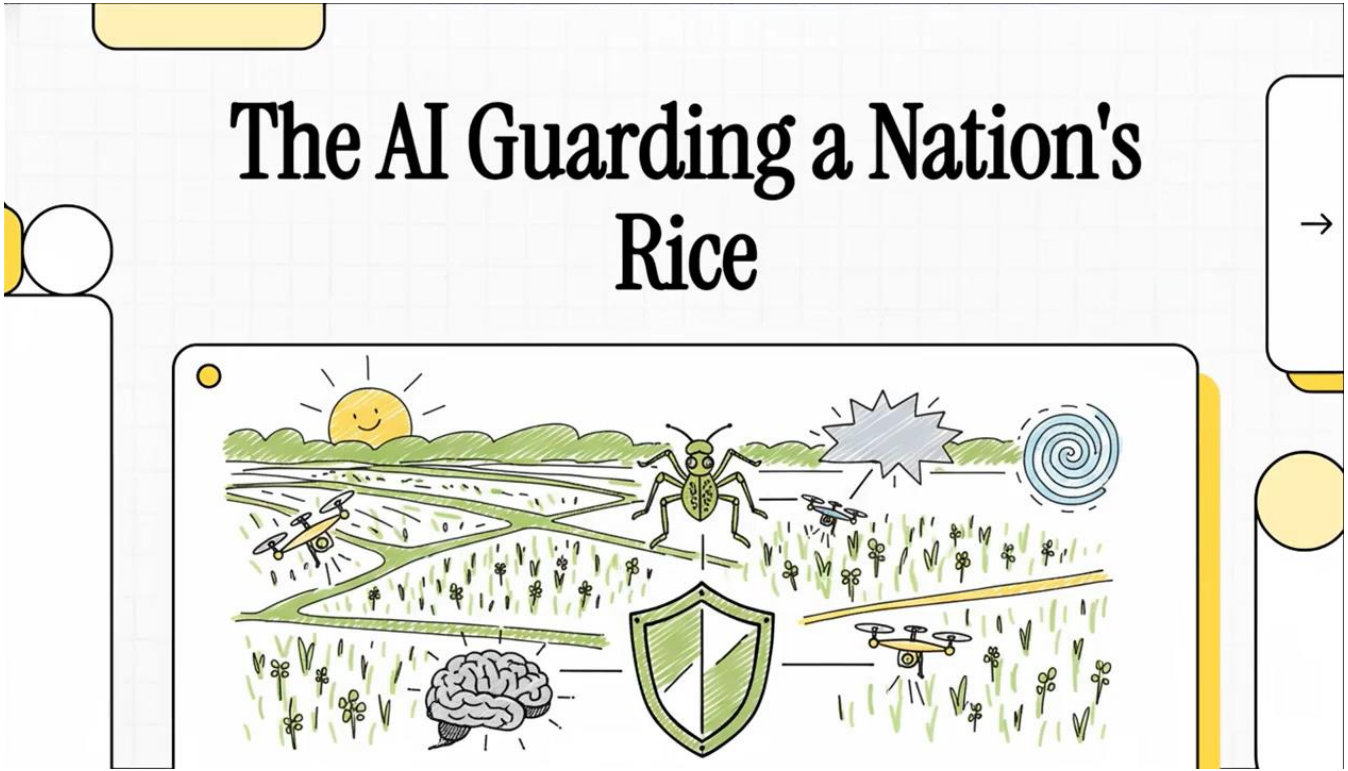
- Rice importance in Sri Lanka: feeds 21M people; 1.8M households rely on paddy farming.
- Challenges: diseases like Bacterial Blight, Rice Blast, Brown Spot can destroy 30% - 80% of harvest; limited access to experts.
- Existing tools limitations: mostly image-based, English-only, not farmer-friendly, require good cameras.

PaddyGuard AI solution:

- Voice & image-based disease detection.
- Treatment recommendations & pest identification.
- Sinhala-compatible advisory chatbot.
- Easy-to-use, mobile-friendly platform.
- Impact: supports SDG 2 (Zero Hunger) and SDG 9 (Innovation & Infrastructure) by protecting crops and empowering farmers.



Problem Definition



PaddyGuard AI: A Multimodal AI Based System for Rice Leaf Disease And Pest Detection, Treatment Recommendation And Knowledge Assistance

Comments for the Proposal and How the Team Addressed Them

Comment Received	How We Addressed It	Status
Expand dataset with more complex and regional dialect data	Created 1,200+ row Sinhala symptom dataset covering 5 regional accents (Southern, Uva, North Central, Sabaragamuwa, Central) with expert-validated agricultural vocabulary.	In Progress
Strengthen model comparison and baseline justification	Conducted 6-model comparative study (NB, LR, RF, SVM, DistilBERT, BERT). SVM selected as best with 96.67% F1 — justified with size and speed trade-off analysis.	Resolved
For voice, collect a verified dataset	Compiled a verified voice dataset with native Sinhala speaker recordings, validated by agricultural domain experts to ensure pronunciation accuracy and disease-terminology correctness across regional dialects.	Resolved
Don't use all the diseases and pest available — limit the scope	Narrowed the system scope to the most prevalent and high-impact rice diseases in Sri Lanka (e.g., Blast, BLB, Brown Spot, Sheath Rot), ensuring higher model accuracy and more focused training data.	Resolved
Use software engineering principles when developing the project	Adopted formal SDLC practices including requirements specification, UML diagrams, modular architecture, version control (Git), code reviews, and unit/integration testing throughout development.	Resolved
Integrate voice component with treatment advisory system	FastAPI /diagnose endpoint returns disease + confidence to Component 3 (Treatment) via shared backend API contract. Integration tested end-to-end.	In Progress

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Problem & Impact / Gravity

30 – 80%	crop loss from rice diseases when untreated
1.8M	paddy farming households in Sri Lanka affected
21M	people fed by Sri Lankan rice production
0	existing tools supporting Sinhala voice input

Who Benefits

Smallholder paddy farmers · Agricultural extension officers · Department of Agriculture Sri Lanka · Agri-input companies · Crop insurance providers

Design Excellence of the Overall Solution

C1 — Voice NLP

Sinhala ASR + 6-model comparison. SVM 96.67% F1. First agricultural Sinhala voice-NLP pipeline with 3-signal OOD detection.

C2 — Leaf Image

CNN multi-architecture comparison (ResNet50, EfficientNetB3, DenseNet121). 92%+ accuracy with Grad-CAM explainability + ODIN OOD.

C3 — Treatment

RAG-based chatbot (IRRI + FAO knowledge) with rule-based advisory engine. Sinhala language treatment delivery.

C4 — Pest Detect

YOLO v8 + CNN (DenseNet121 best) for 5 pest classes. OOD detection for unknown insects.

Voice-Based Rice Disease Diagnosis Using Speech Recognition And Natural Language Processing



IT22273680 | Weerasinghe K G J P

Contribution of your component to the overall solution

Enables low-literacy Sinhala-speaking paddy farmers to voice-diagnose rice diseases without cameras or technical expertise, using the world's first Sinhala agricultural voice-NLP pipeline integrated with adaptive follow-up questioning and OOD detection.

Novelty of your component

First Sinhala voice-based rice disease NLP system · 6-model comparison (SVM vs. BERT vs. DistilBERT) · 3-signal OOD detection · Adaptive confidence-threshold follow-up questioning · Expert-validated Sinhala regional dialect dataset.

Voice-Based Rice Disease Diagnosis — 50% Progress & Prototype

How the Solution Addresses the Sub-Problem

Sinhala Voice Input
Whisper-small fine-tuned on Sinhala transcribes farmer voice to text — no English or literacy required.

Automatic Translation
Google Translate converts Sinhala text to English, feeding a domain-specific SVM classifier.

Disease Classification
SVM (TF-IDF) achieves 96.67% F1 across 4 classes — outperforms BERT in this short keyword-rich domain.

Adaptive Follow-up
When confidence < 0.70, a rule-based dialogue asks up to 3 targeted questions to refine the diagnosis.

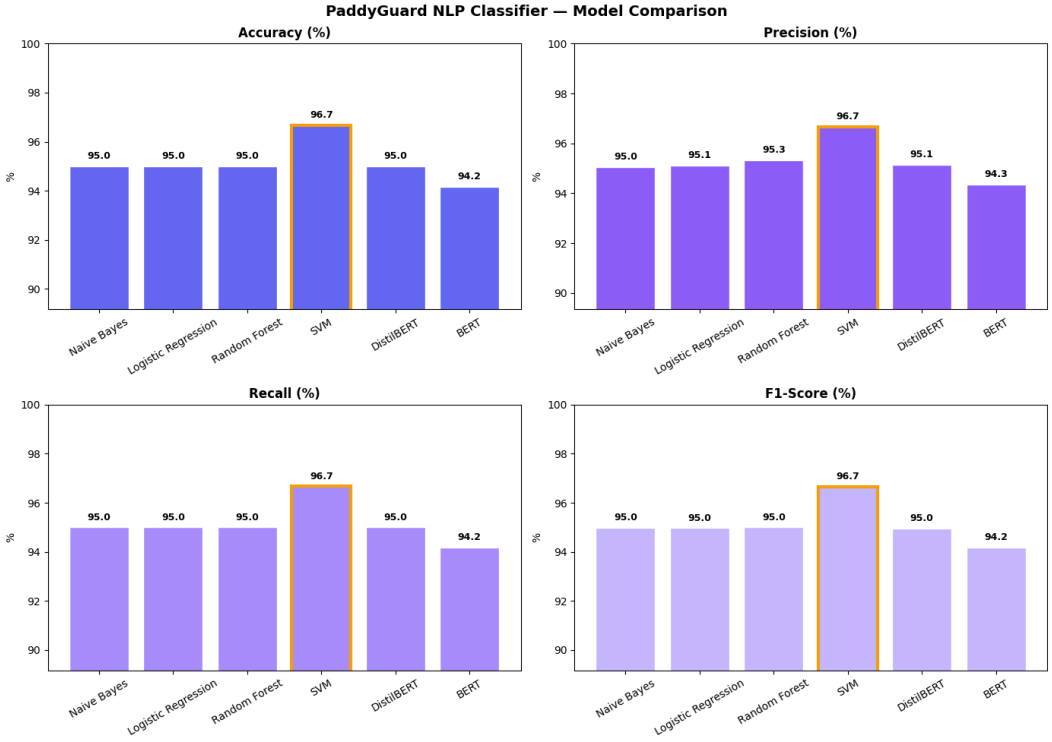
OOD Detection
3-signal method (vocabulary + confidence + entropy) rejects non-disease inputs without retraining.

User Requirements Addressed

FR1	**Voice input in Sinhala — no typing**	Whisper ASR live on mobile
FR2	**Works on basic smartphones**	Browser-based, no app install
FR3	**Returns result in Sinhala**	Sinhala text + TTS response
FR4	**Handles unclear symptom descriptions**	Follow-up question module (max 3 Q)
FR5	**Rejects unrelated voice input**	3-signal OOD detection
FR6	**Connects to treatment component**	FastAPI /diagnose API integrated

Voice-Based Rice Disease Diagnosis — Design Excellence & User Feedback

Design Excellence — 6-Model Comparative Study



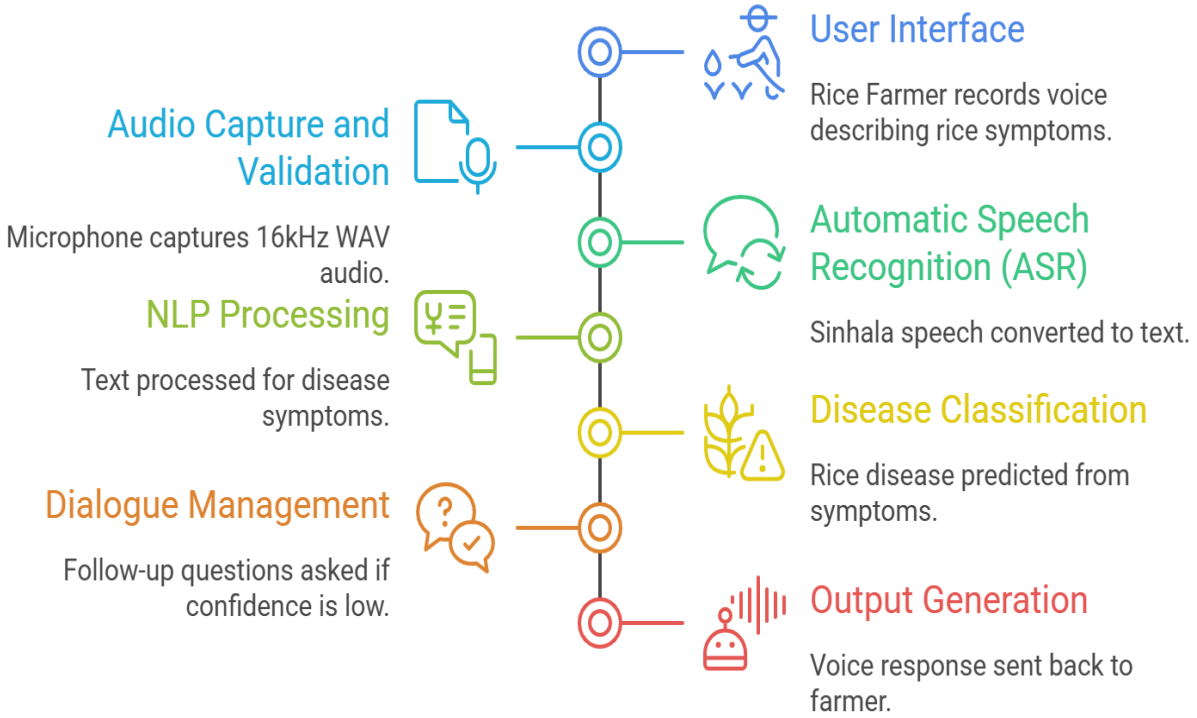
OOD Detection — 3-Signal Framework

Signal 1	**Vocabulary Check** — Instantly rejects inputs with zero symptom words (e.g. "ගෙදර යන්න")
Signal 2	**Confidence Threshold** — Max class probability < 0.50 → OOD (prevents SVM over-confidence)
Signal 3	**Entropy Check** — Prediction entropy > 1.20 (base 2) → spread probabilities → OOD

User Feedback on Prototype

- Sinhala microphone interface was intuitive — no instruction needed.
- Sinhala script results were understood by all farmers immediately.
- Confidence score % increased farmer trust in 'Confident' results.
- Uva/NC accents caused ASR errors — regional dataset expansion planned.
- Farmers requested voice output — gTTS si-LK TTS planned next sprint.

System Overview And Dependencies



Rice Leaf Disease Classification with Out-of-Distribution Detection

Hewanayake H.M.L.M | IT22168740



Contribution to Overall Solution

Classifies rice leaf diseases from images, filters out invalid inputs, and explains predictions visually.

Novelty

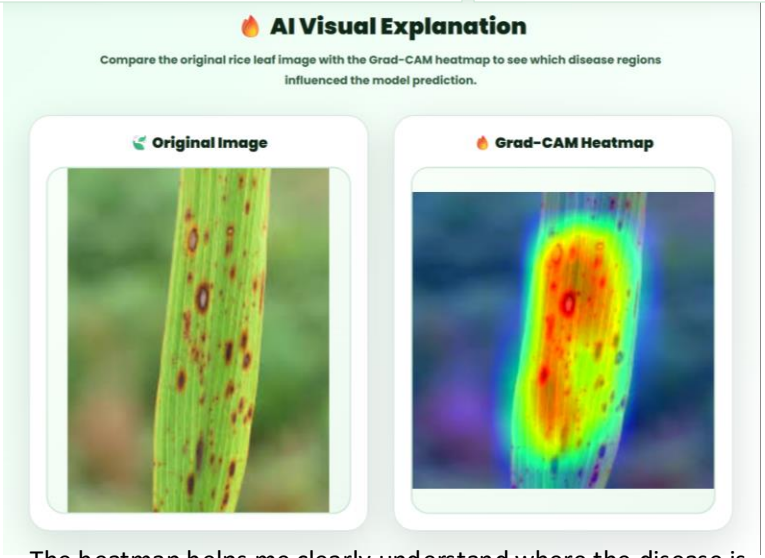
- Integrated OOD detection— rejects non-rice/unknown images
- Grad-CAM visual explanations highlighting disease lesion regions

Rice Leaf Disease Classification with OOD Detection

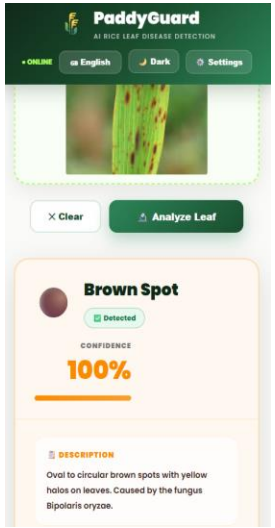
Sub-Problem	Solution
Wrong predictions	OOD Detection — Confidence < 0.78 → rejects image with "Unknown image" message
Low farmer trust	Grad-CAM Heatmap — Red overlay on disease shows exact lesion location
Difficult upload	Simple Interface — Camera / Gallery upload with real-time preview

0.78

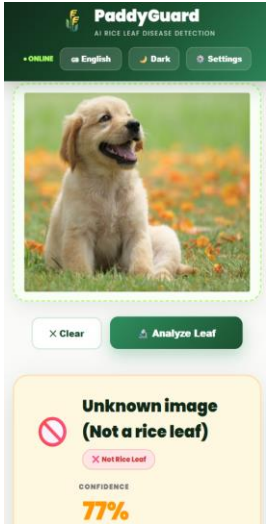
OOD Threshold



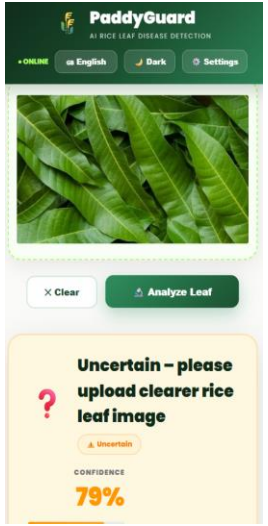
The heatmap helps me clearly understand where the disease is located. I can trust this app for making decisions in my field.



Correct Detection



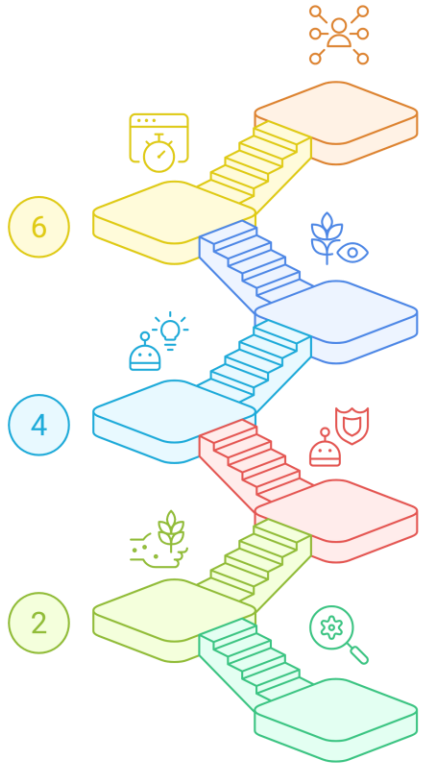
ODD Rejection



ODD Rejection

Rice Leaf Disease Classification with OOD Detection

User Requirements Addressed



7 Usability Design
Creating an intuitive interface for low literacy users.

6 Performance Optimization
Achieving inference times under 500ms.

5 Explainability
Visualizing model decisions with Grad-CAM heatmaps.

4 Confidence Display
Showing prediction confidence percentages to users.

3 OOD Detection
Rejecting unknown images with an 87.3% rejection rate.

2 Disease Classification
Accurately identifying rice diseases with a 92.34% accuracy.

1 Image Preprocessing
Resizing and normalizing images for model input.

Design Excellence

Characteristic	Accuracy/Score
EfficientNetB3 Transfer Learning	97.61% test accuracy, 97.61% Macro F1-score
Confidence-based OOD Rejection	91.86% OOD system accuracy
Farmer-friendly Prediction System	0.78 → Unknown, 0.78–0.90 → Uncertain, > 0.90 → Final prediction
Grad-CAM Heatmaps	Improves transparency and farmer trust
Real-world Robustness	Uses augmentation and preprocessing
Lightweight Flask API	500ms inference time

USER FEEDBACK

Smallholder Farmer – Crop Monitoring Experience

"The heatmap helps me clearly understand where the disease is located. I can trust this app for making decisions in my field."

Made with Napkin

A Multimodal AI System for Rice Pest Detection With Out-of-Distribution Detection



IT22065858 | KALHARA E P K

Contribution of component to the overall solution

Automatic detection of rice pests from images, helping farmers identify pest infestations early.

Novelty of component

Integration of deep learning–based pest detection with Out-of-Distribution (OOD) detection to identify.

Problem Definition & Research Objective

How the Solution Addresses the Sub-Problem

- Farmers can upload pest images through the web.
- The system preprocesses images.
- Deep learning CNN models analyze the uploaded image and identify rice pests.
- YOLO-based detection is used for real-time pest.
- OOD detection identifies unknown insect images and avoids incorrect predictions.

User Requirements Addressed by the Solution

- Upload rice pest images.
- Detect common rice pests.
- Display prediction with confidence score.
- Detect unknown insects using OOD.
- Provide treatment guidance.
- Easy-to-use interface.

Target Pests classes : • Brown Planthopper • Stem Borer • Leaf Folder • Rice Hispa • Gall Midge

Methodology & Latest Technologies

Design Excellence / Contribution

- Compared and evaluated 5 **CNN architectures**: DenseNet121 , MobileNetV2 , EfficientNet , InceptionV3 , ResNet50

- DenseNet121 achieved the best overall performance.
- Added **Out-of-Distribution (OOD)** detection to improve reliability.
- Designed using **FastAPI** backend and React frontend.
- Supports real-world agricultural environments with varying image conditions.
- Supports early pest detection and smart farming.

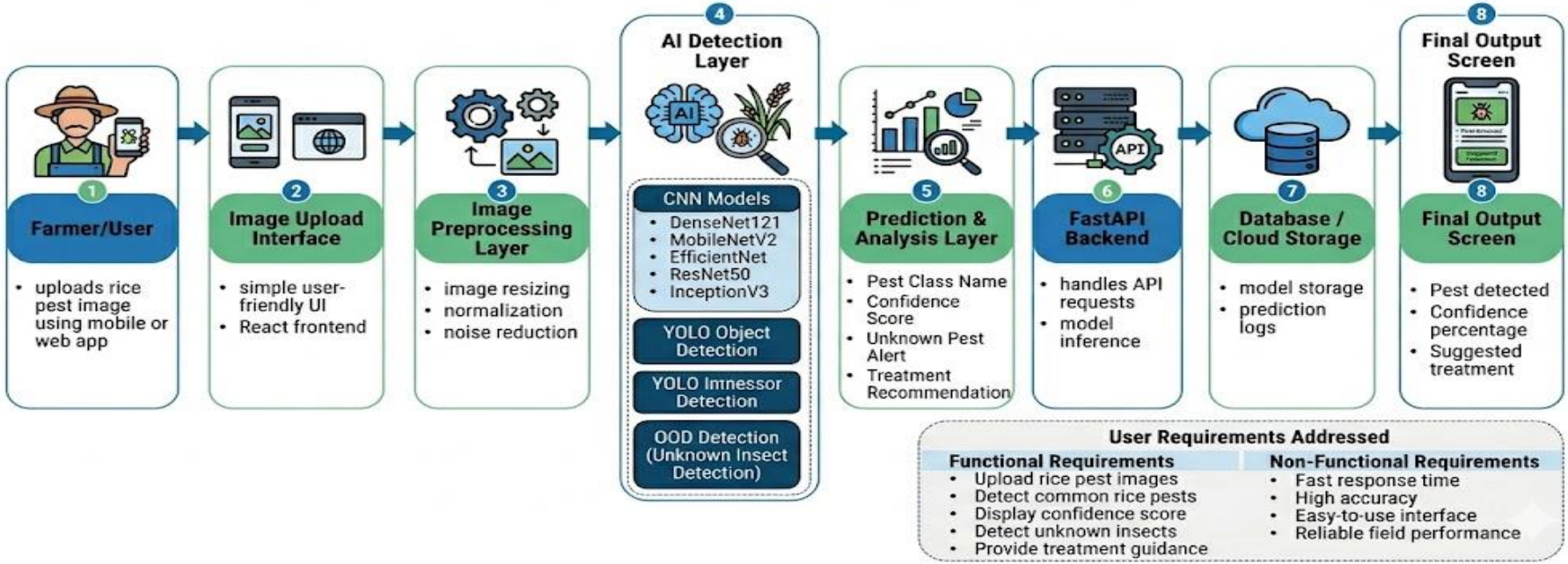
User Feedback on Prototype

- Easy image upload process
- Fast prediction response
- Helpful confidence score display

Improvements Suggested

- Improve detection for blurry field images
- Add more rice pest categories
- Develop mobile application version

Rice Pests detection system workflow



Treatment Advisory System and Agricultural Chatbot

Keshan B K | IT22303820



Contribution to Overall Solution

Provides intelligent agricultural treatment recommendations through a multilingual conversational AI system using:

- NLP-based query understanding
- RAG-based knowledge retrieval
- Context-aware conversation memory
- Explainable AI response generation

Novelty

- Conversational AI for paddy disease & pest treatment guidance
- RAG-based agricultural knowledge retrieval
- Sinhala + English multilingual chatbot support
- Context-aware adaptive memory system
- Confidence-aware AI recommendations
- Intelligent follow-up question generation

Treatment Advisory System and Agricultural Chatbot

Treatment Advisory System and Agricultural Chatbot

Sub-Problem	AI-Based Solution
Farmer Communication Difficutties	Multilingual conversational AI with Tintata + English NLP
Incovrect Treatrhent Guidance	Confidence aware explainable AI recommendations
Lack of Agricultural Centext	Contest aware memory + RAG knowledee retrieral
Low Interoction Quality	Intelligent follow-up duestion generatiori

- ✓ Multtlingual NLP
- ✓ RAG Retrieval
- ✓ Explalnable Al
- ✓ Adaptive Meimory
- ✓ Adaptive Memory
- ✓ Follow-up Intelligence
- ✓ Confidence Scoring

Core Al Pipeline



Treatment Advisory System and Agricultural Chatbot

USER REQUIREMENTS ADDRESSED

1. **Query Processing**
   Normalizes and processes multilingual farmer queries

2. **Intent Detection**
   Identifies:
   - disease_detection
   - pest_detection
   - treatment_recommendation
   - prevention_advice

3. **Semantic Retrieval**
   Retrieves relevant disease, pest, and FAQ knowledge

4. **Confidence Analysis**
   Measures:
   - semantic confidence
   - entity confidence
   - RAG confidence
   - conversation confidence

5. **Contextual Memory**
   Maintains:
   - entity continuity
   - intent continuity
   - semantic memory

6. **Follow-up Intelligence**
   Generates clarification questions for low-confidence queries

7. **Explainable AI**
   Provides:
   - transparent responses
   - confidence-aware guidance
   - safe agricultural recommendations

RESEARCH FEATURES

- **Conversational Agricultural AI**
  Natural farmer interaction support
- **RAG-Based Intelligence**
  Context-enhanced response generation
- **Adaptive Memory**
  Context-aware multi-turn conversations
- **Explainable AI**
  Confidence-aware advisory recommendations
- **Explainable AI**
  Confidence-aware advisory recommendations

User Feedback

Smallholder Farmer – Crop Monitoring Experience

"The Chatbot helps me code understand the treatment process clearly in Sinhala. I can ask question naturally and receive useful farming advice."

Market, Value Proposition And Commercialization

Value Proposition

- Farmers: Instant Sinhala voice-based disease diagnosis & treatment advice.
- Government: Scalable digital advisory for rural farmers.
- Agri Companies: Targeted product recommendations at diagnosis stage.
- Society: Reduced crop losses and improved food security.

Target Market

- Primary: Department of Agriculture & extension services.
- Secondary: Agricultural cooperatives, crop insurance, agri-input companies.
- End Users: ~1.8M paddy farming households in Sri Lanka.

Commercialization

- Government licensing as national agricultural advisory tool.
- API integration for agri-input companies.
- Grants & SaaS analytics for cooperatives and regional expansion.



Thank You